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(54) **FREE PLAY WAGERING GAME
REDEMPTION OF CRYPTOGRAPHIC
TOKENS FOR MONETARY AWARDS**

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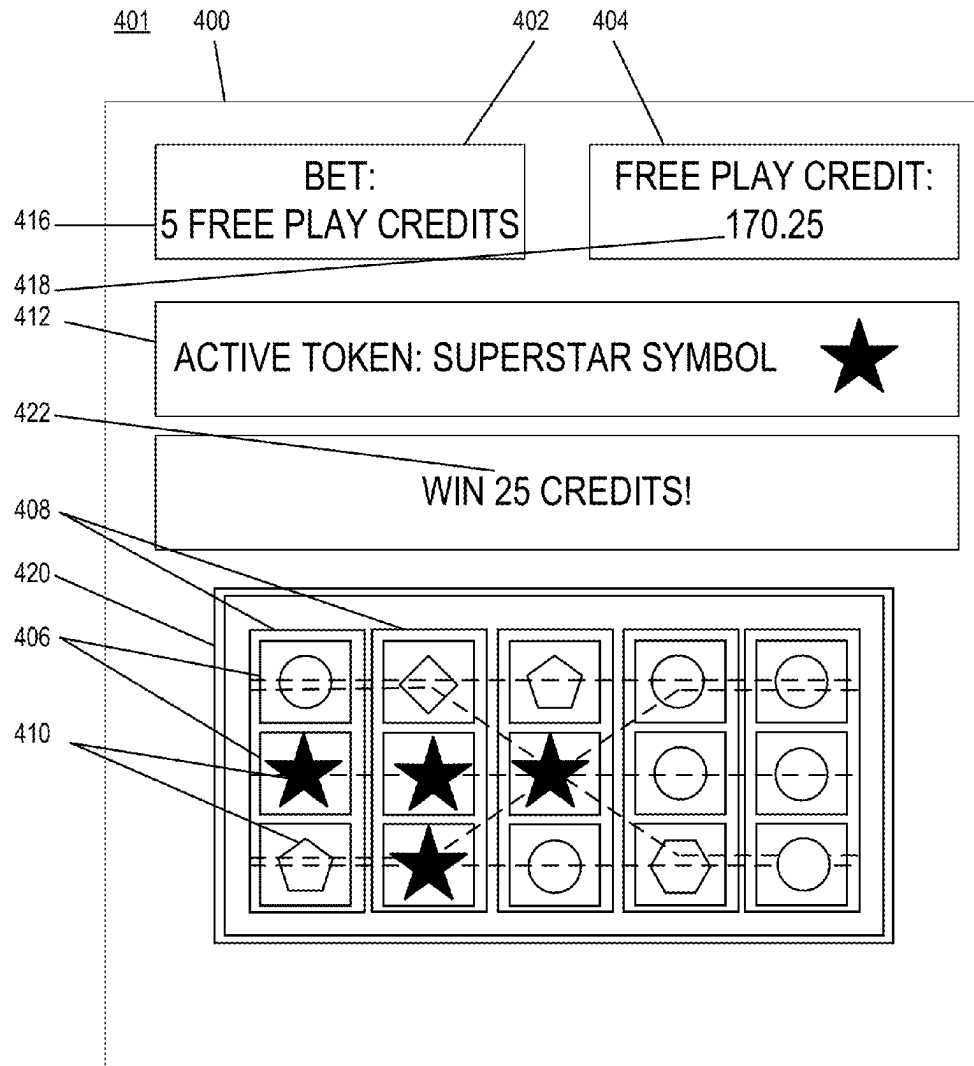
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(57) **ABSTRACT**

A method includes executing a plurality of games of a wagering game. Execution of each game includes deducting a non-monetary wager amount from a non-monetary credit balance amount, determining a result for the wagering game, and, based on the result being a winning result and the non-monetary wager amount, adding a non-monetary award amount to the non-monetary credit balance amount. The method further includes, after executing the plurality of games, receiving an instruction from a player of the wagering game to redeem a cryptographic token to award a portion of the non-monetary credit balance amount to the player as a monetary award amount. The method further includes determining, by the processor circuit, the monetary award amount based on the non-monetary credit balance amount and the cryptographic token. The method further includes, based on redeeming the cryptographic token, awarding, by the gaming device, the monetary award amount to the player.



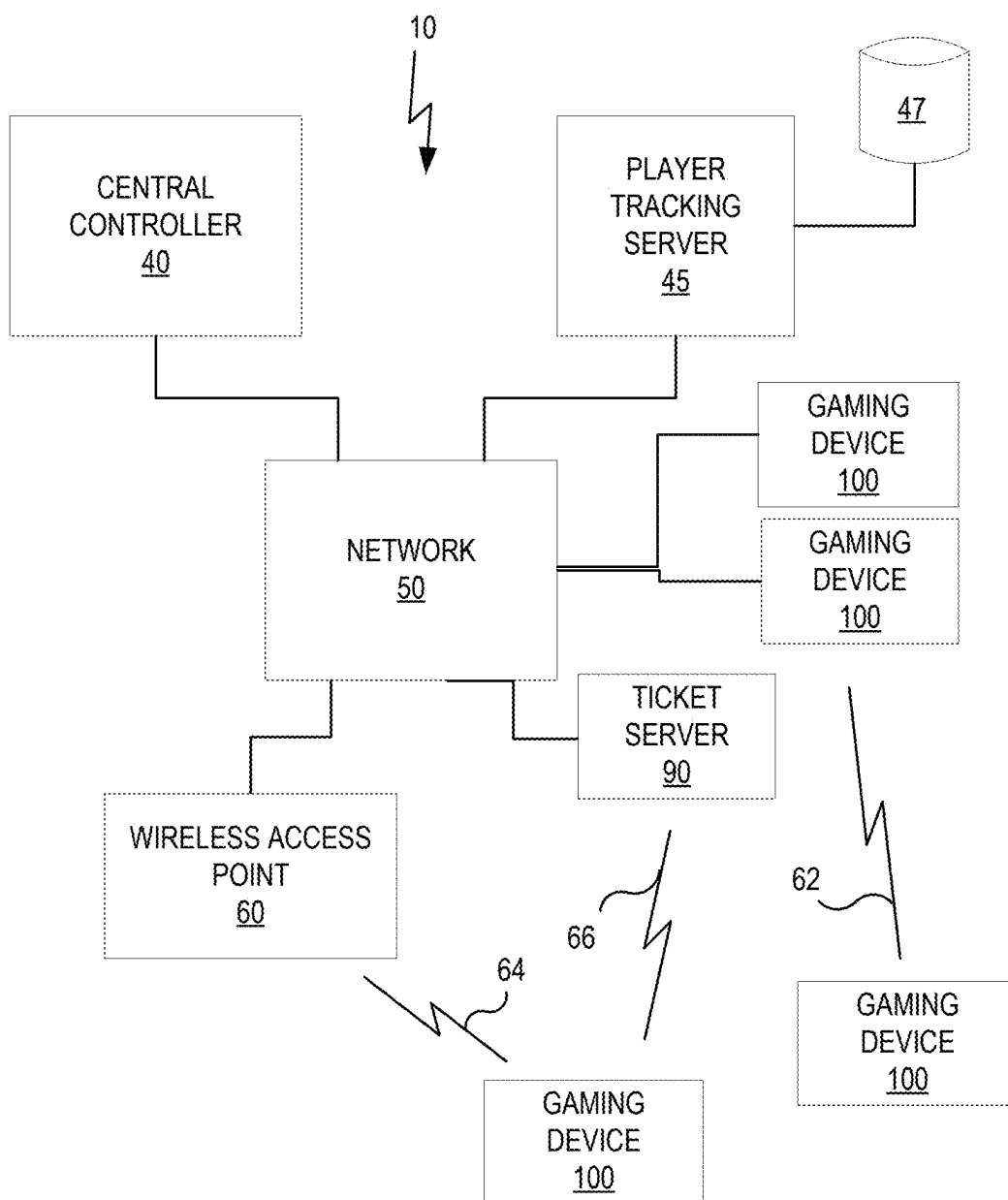


FIG. 1

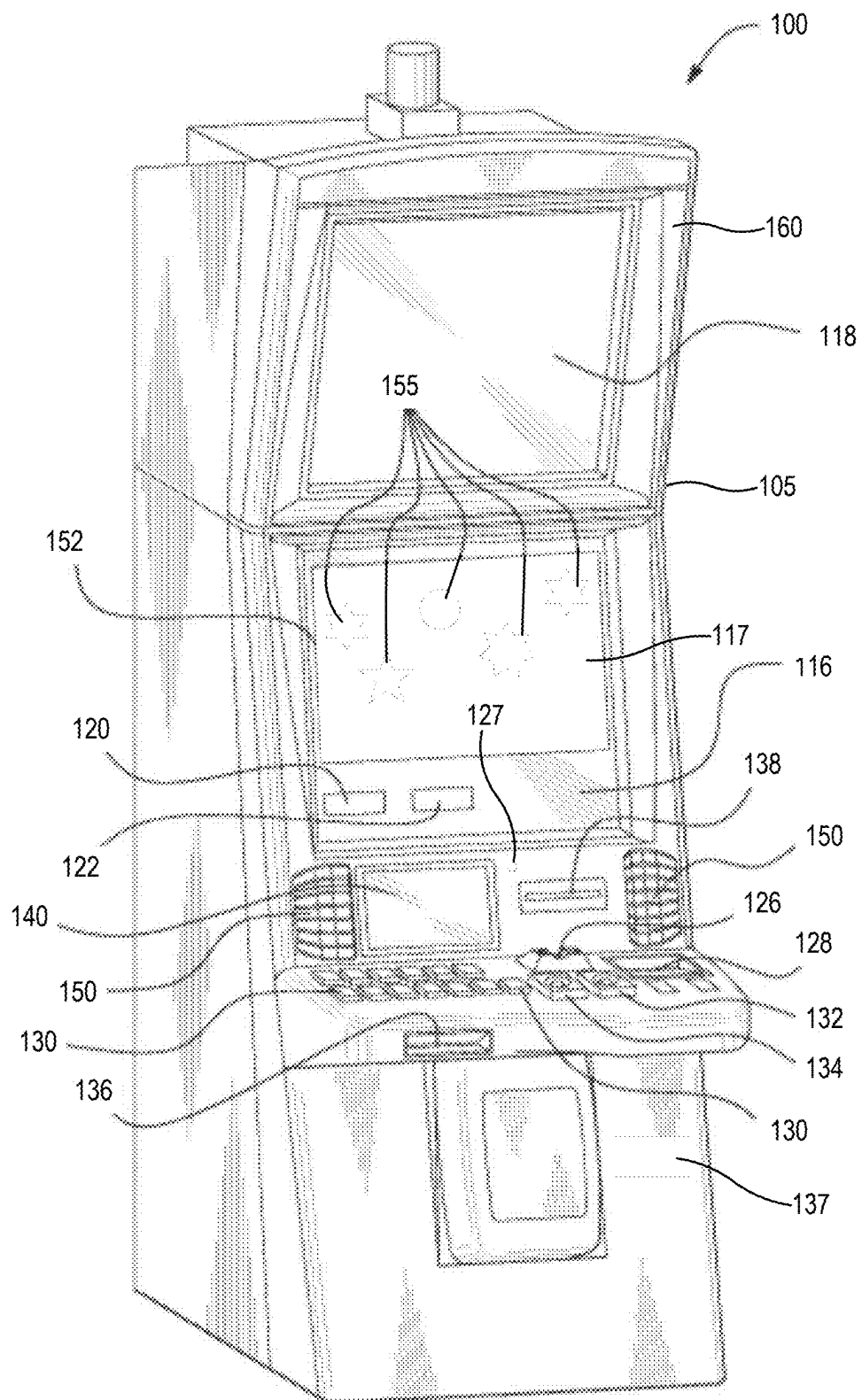


FIG. 2A

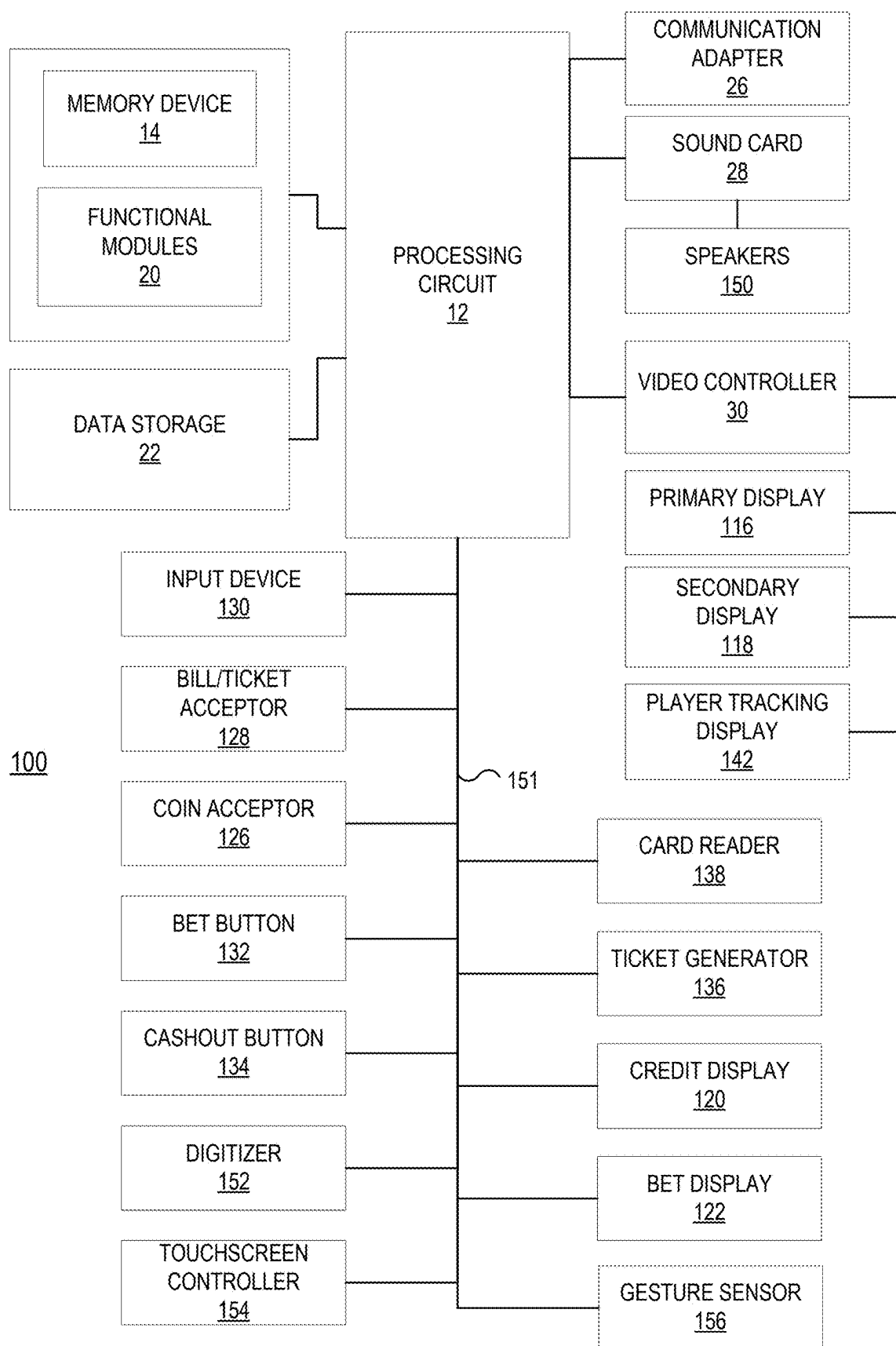


FIG. 2B

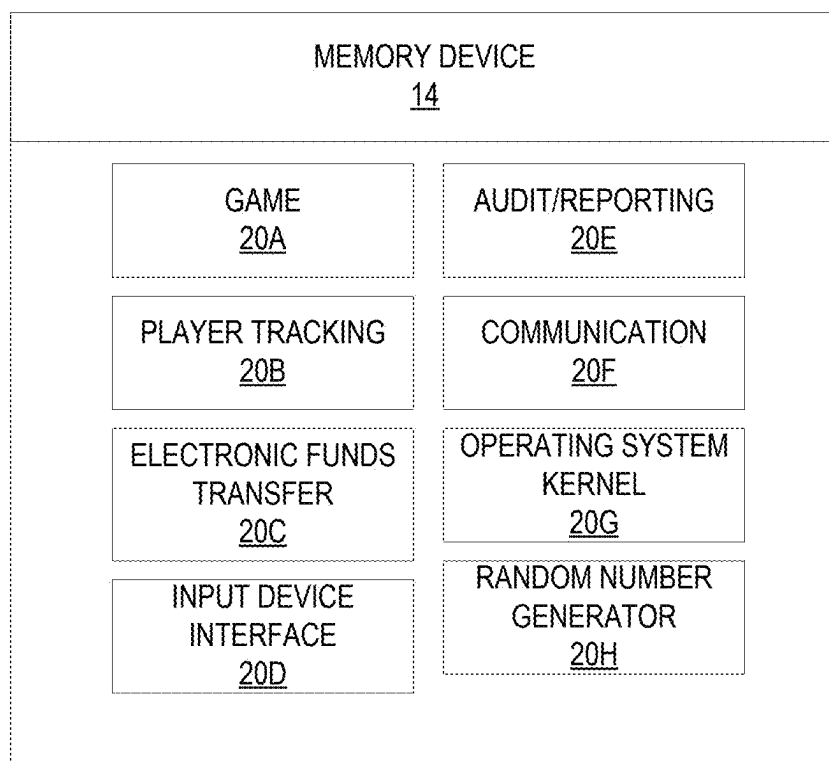


FIG. 2C

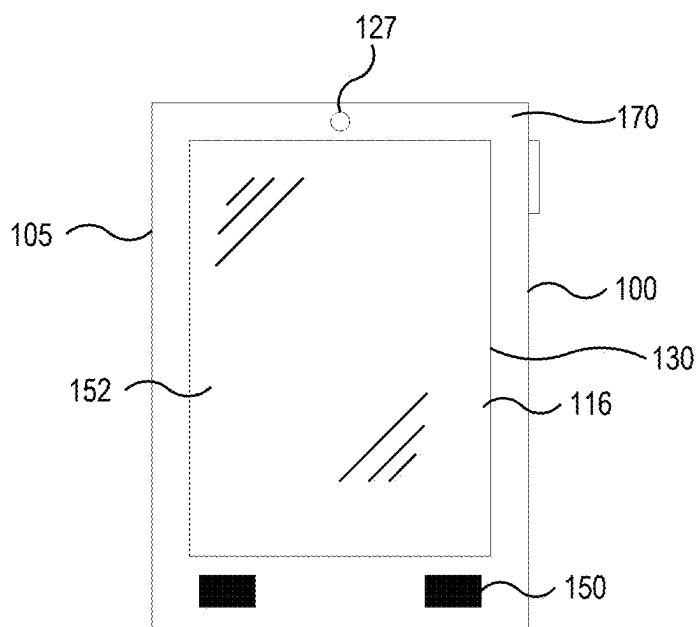


FIG. 2D

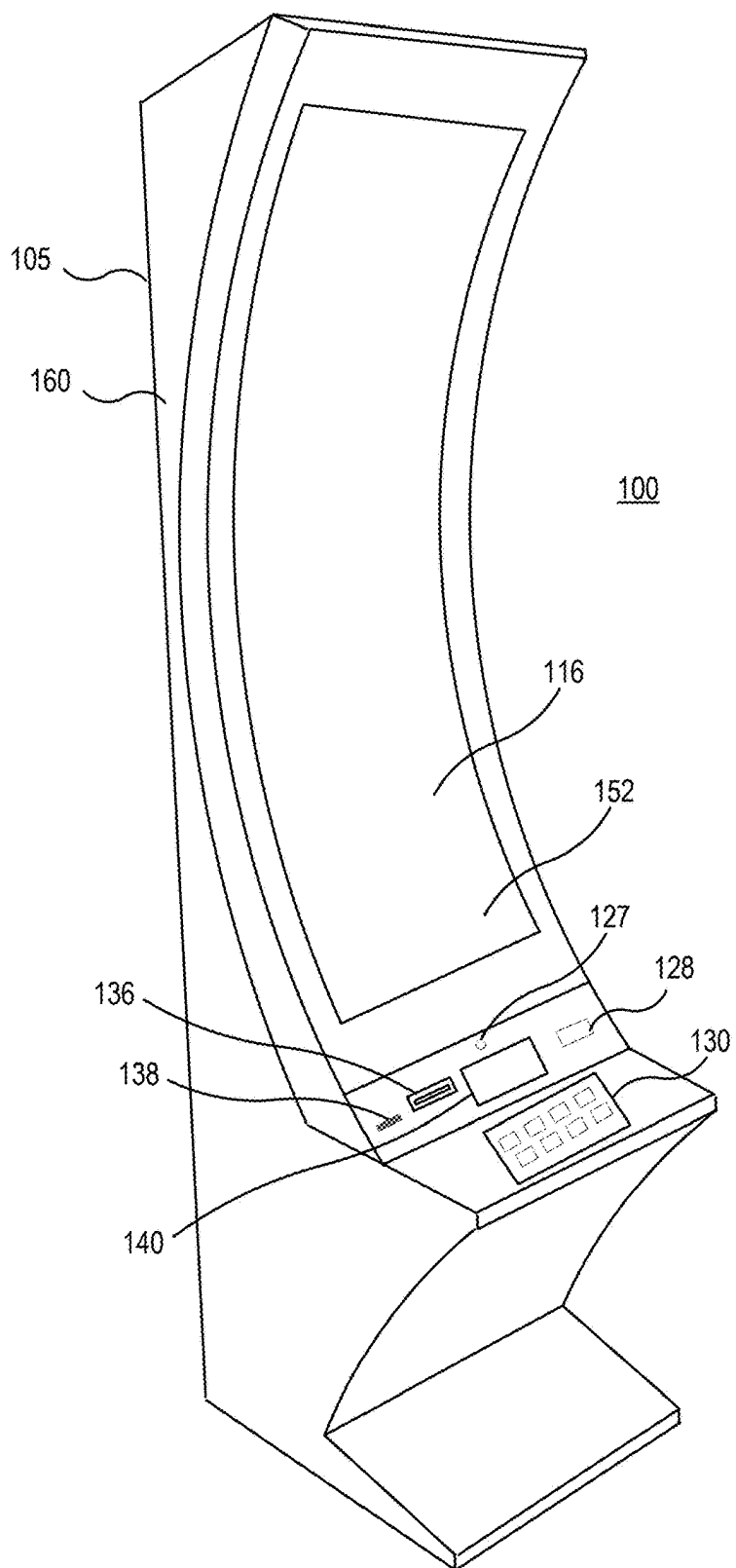


FIG. 2E

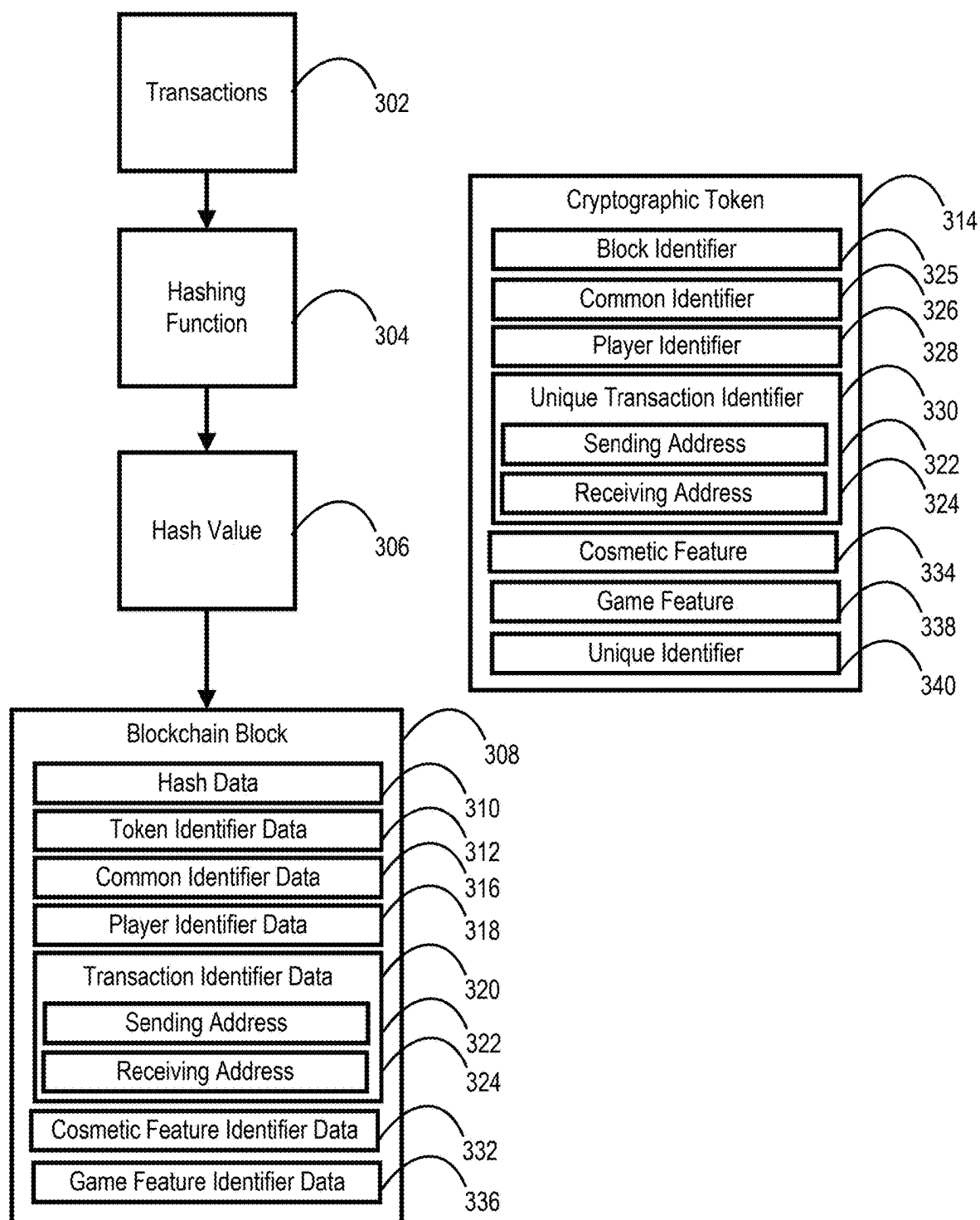


FIG. 3

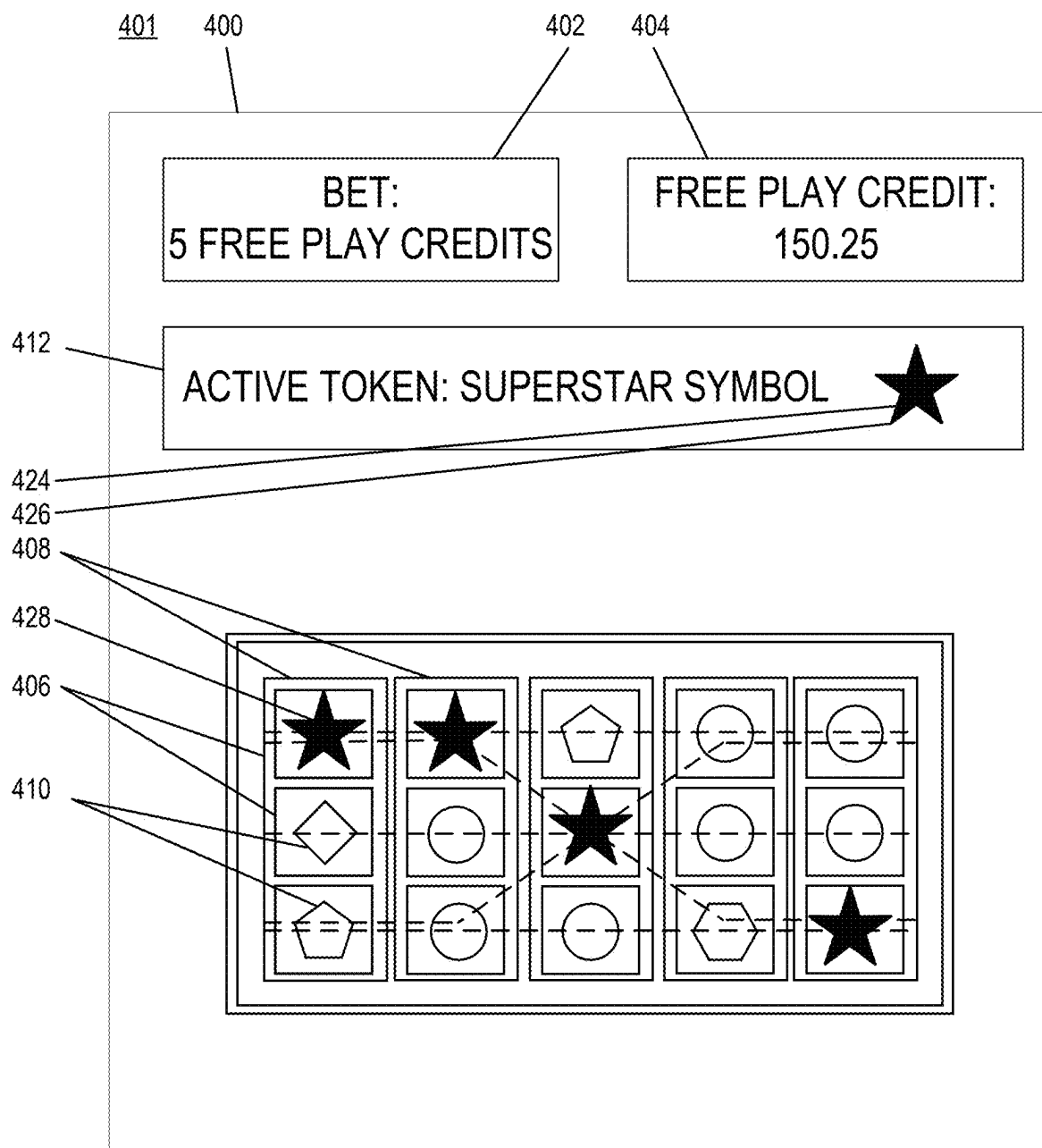


FIG. 4A

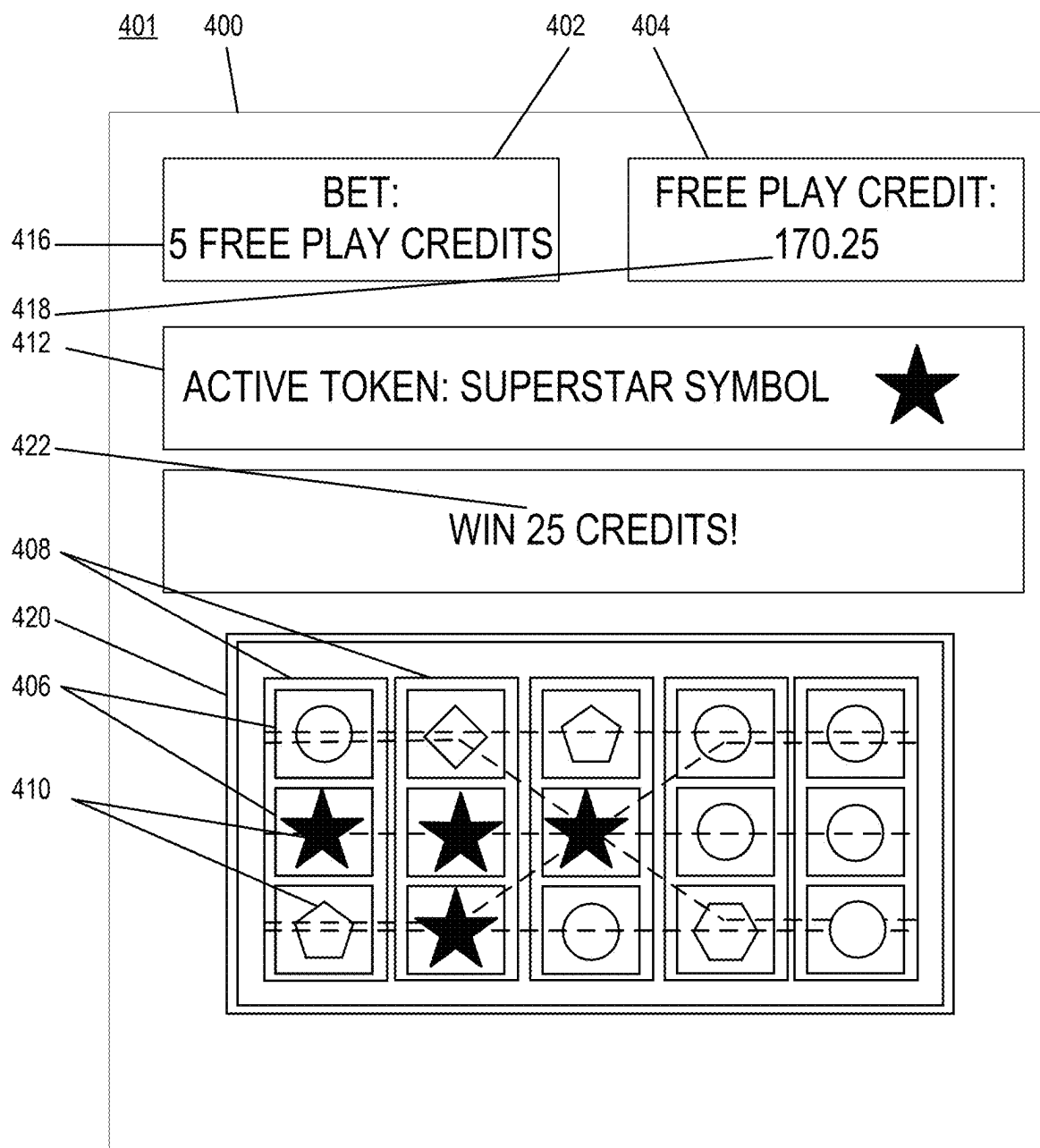


FIG. 4B

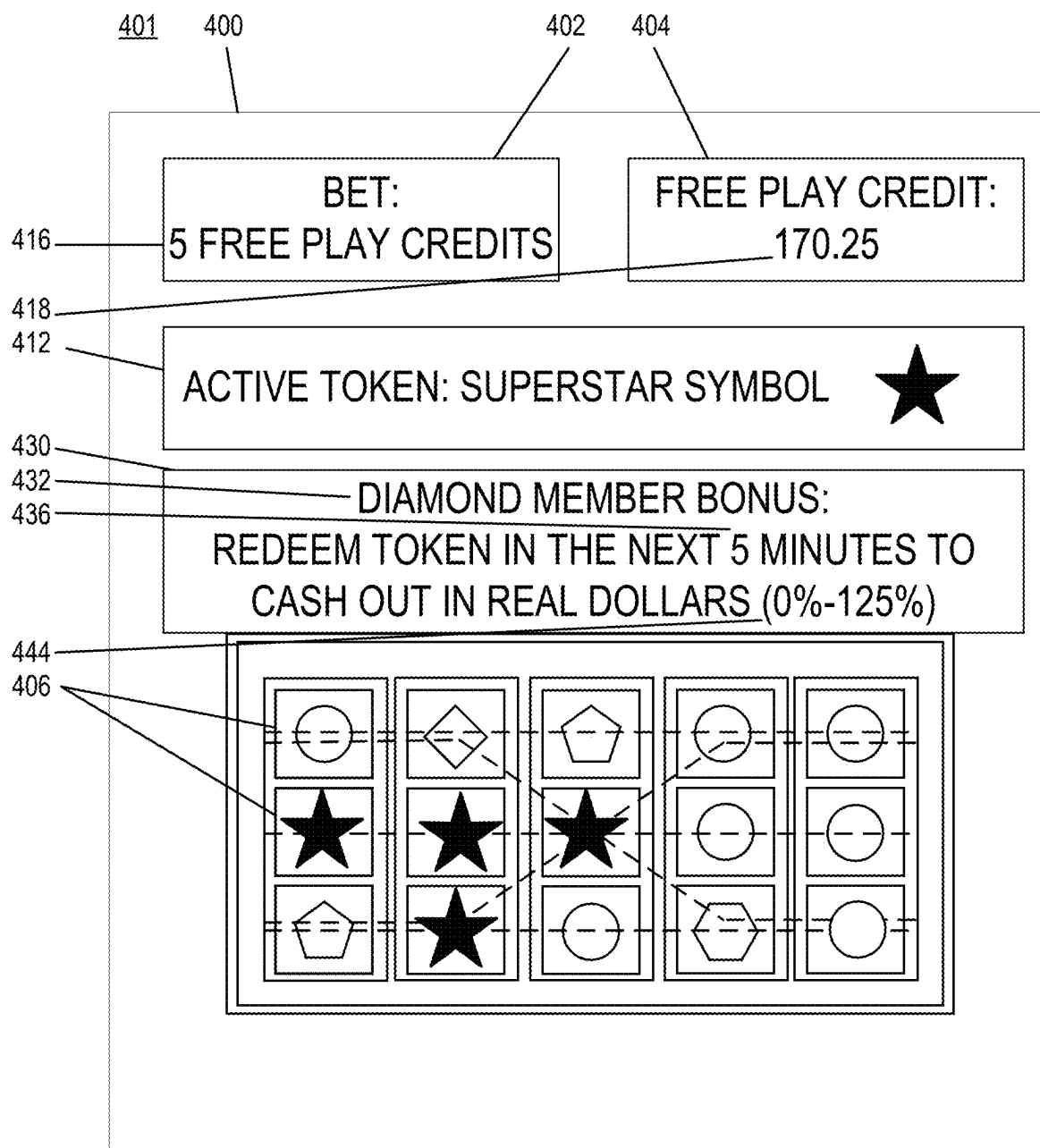


FIG. 4C

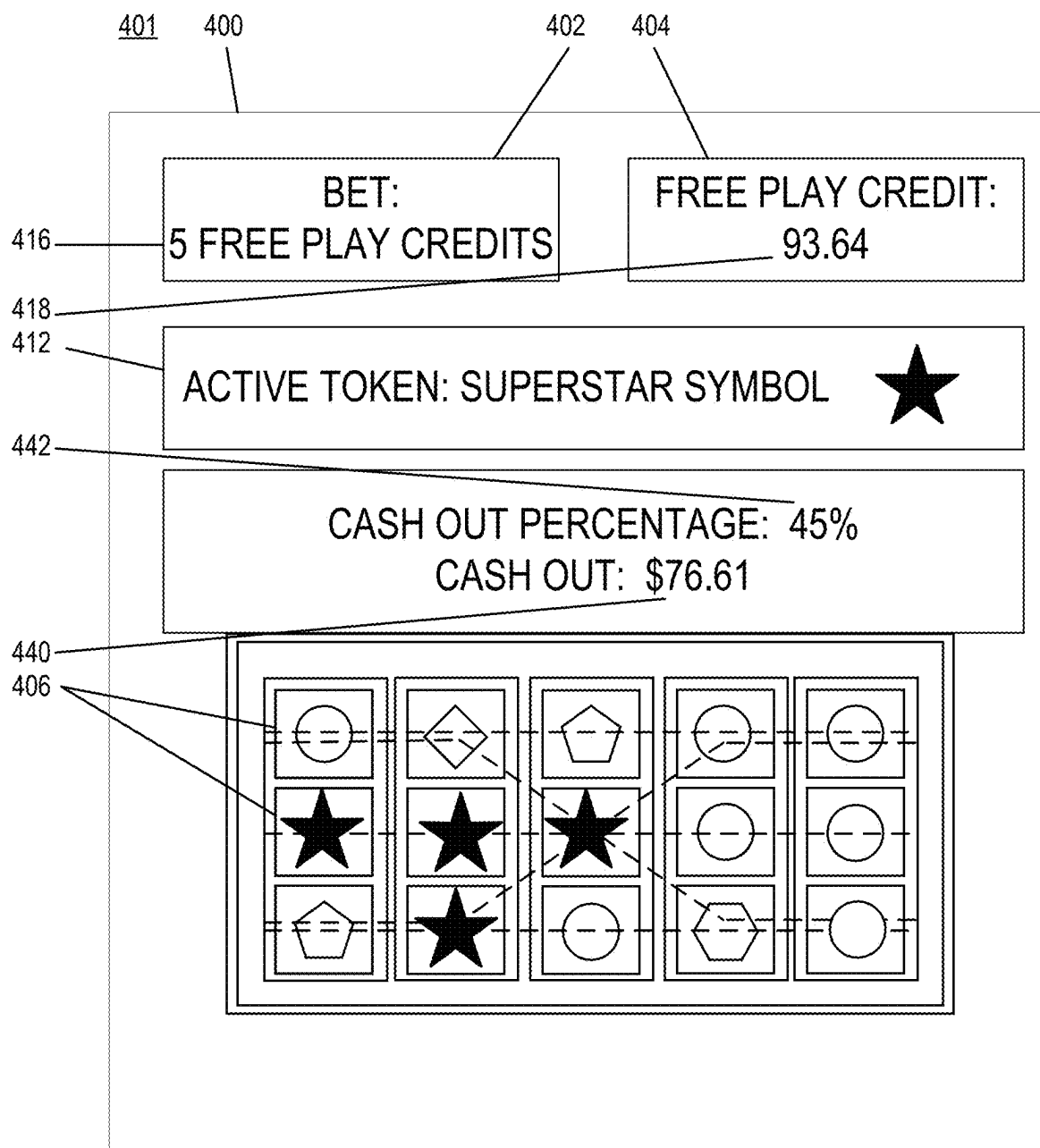


FIG. 4D

500

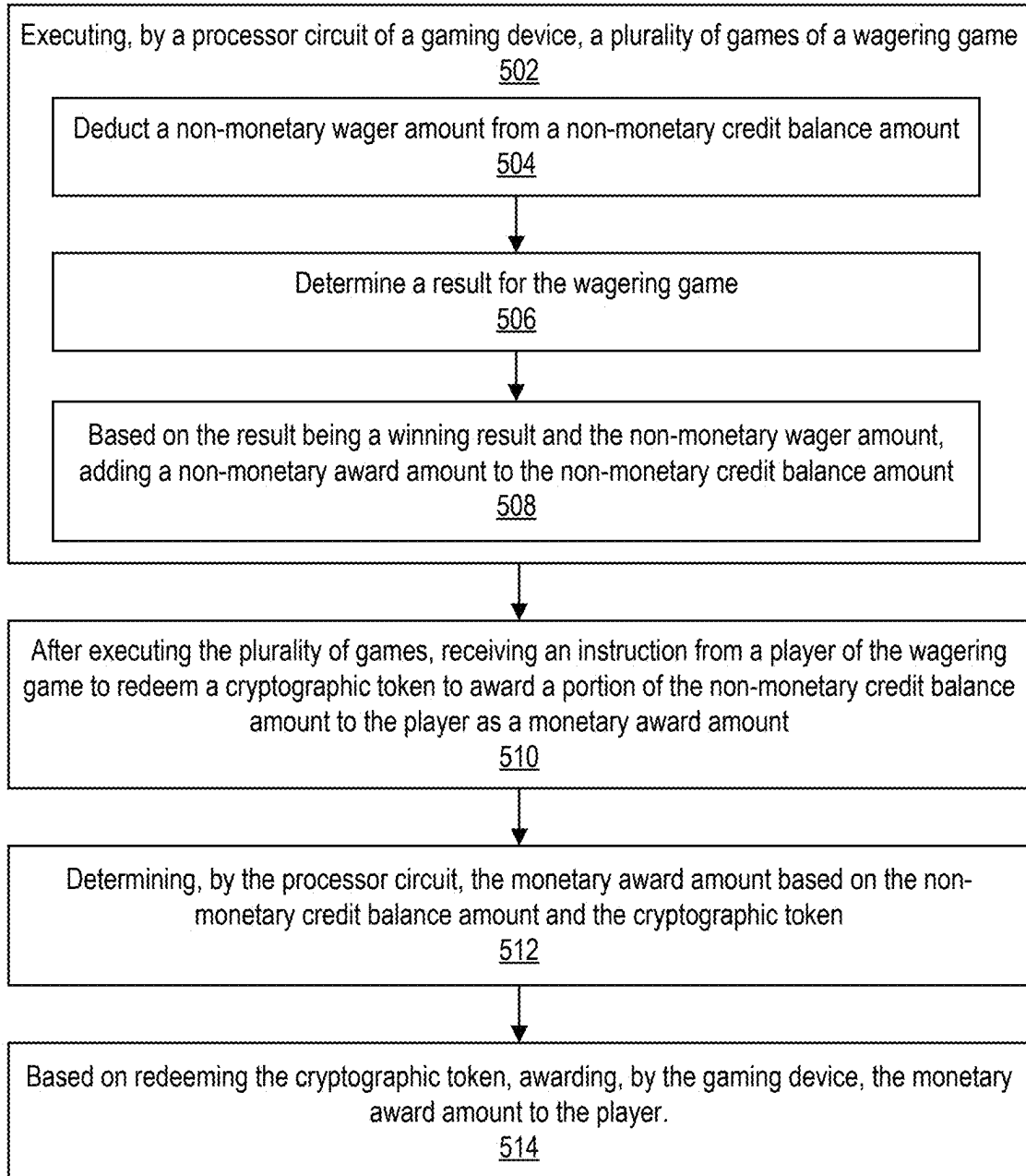


FIG. 5

FREE PLAY WAGERING GAME REDEMPTION OF CRYPTOGRAPHIC TOKENS FOR MONETARY AWARDS

BACKGROUND

[0001] Embodiments described herein relate to game play features with wagering games, and in particular to free-play wagering game redemption of cryptographic tokens for monetary awards in a gaming environment, such as in a casino environment, and related devices, systems, and methods. Some wagering games, such as slot games and/or video poker games provided at Electronic Gaming Machines (EGMs) and/or mechanical gaming devices in a casino environment, may include unique game features as an additional incentive for play of a wagering game. There is a need address these and other technical problems for providing additional options for player interaction and enjoyment to encourage wagering game play.

BRIEF SUMMARY

[0002] According to some embodiments, a system includes a processor circuit and a memory coupled to the processor circuit. The memory includes machine-readable instructions that, when executed by the processor circuit, cause the processor circuit to execute a plurality of games of a wagering game. Execution of each game causes the processor circuit to deduct a non-monetary wager amount from a non-monetary credit balance amount, determine a result for the wagering game, and, based on the result being a winning result and the non-monetary wager amount, add a non-monetary award amount to the non-monetary credit balance amount. The instructions further cause the processor circuit to, after the execution of the plurality of games, receive an instruction from a player of the wagering game to redeem a cryptographic token to award a portion of the non-monetary credit balance amount to the player as a monetary award amount. The instructions further cause the processor circuit to determine the monetary award amount based on the non-monetary credit balance amount and the cryptographic token. The instructions further cause the processor circuit to, based on the redemption of the cryptographic token, award the monetary award amount to the player.

[0003] According to some embodiments, a gaming device includes a display device, an input device, a processor circuit, and a memory coupled to the processor circuit. The memory includes machine-readable instructions that, when executed by the processor circuit, cause the processor circuit to receive, via the input device, a plurality of non-monetary wagers for a wagering game. The instructions further cause the processor circuit to, based on the plurality of non-monetary wagers, execute a plurality of games of a wagering game. Execution of each game causes the processor circuit to deduct a non-monetary wager amount of the non-monetary wager from a non-monetary credit balance amount, display, by the display device, a result for the wagering game, and, based on the result being a winning result and the non-monetary wager amount, add a non-monetary award amount to the non-monetary credit balance amount. The instructions further cause the processor circuit to, after the execution of the plurality of games, receive, at the input device, an instruction from a player of the wagering game to redeem a cryptographic token to award a portion of the

non-monetary credit balance amount to the player as a monetary award amount. The instructions further cause the processor circuit to determine the monetary award amount based on the non-monetary credit balance amount and the cryptographic token. The instructions further cause the processor circuit to, based on the redemption of the cryptographic token, display, by the display device, an indication of the monetary award amount, and award the monetary award amount to the player.

[0004] According to some embodiments, a method includes executing, by a processor circuit of a gaming device, a plurality of games of a wagering game. Execution of each game includes deducting a non-monetary wager amount from a non-monetary credit balance amount, determining a result for the wagering game, and, based on the result being a winning result and the non-monetary wager amount, adding a non-monetary award amount to the non-monetary credit balance amount. The method further includes, after executing the plurality of games, receiving an instruction from a player of the wagering game to redeem a cryptographic token to award a portion of the non-monetary credit balance amount to the player as a monetary award amount. The method further includes determining, by the processor circuit, the monetary award amount based on the non-monetary credit balance amount and the cryptographic token. The method further includes, based on redeeming the cryptographic token, awarding, by the gaming device, the monetary award amount to the player.

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWINGS

[0005] FIG. 1 is a schematic block diagram illustrating a network configuration for a plurality of gaming devices according to some embodiments.

[0006] FIG. 2A is a perspective view of a gaming device that can be configured according to some embodiments.

[0007] FIG. 2B is a schematic block diagram illustrating an electronic configuration for a gaming device according to some embodiments.

[0008] FIG. 2C is a schematic block diagram that illustrates various functional modules of a gaming device according to some embodiments.

[0009] FIG. 2D is perspective view of a gaming device that can be configured according to some embodiments.

[0010] FIG. 2E is a perspective view of a gaming device according to further embodiments.

[0011] FIG. 3 is a schematic block diagram illustrating various operations for a blockchain token transaction according to some embodiments.

[0012] FIGS. 4A-4D illustrate a graphical user interface (GUI) for providing a primary wagering game and a cryptographic token enabled secondary game at a gaming device, according to some embodiments.

[0013] FIG. 5 is a flowchart illustrating operations of systems/methods of a primary wagering game and a cryptographic token enabled secondary game at a gaming device, according to some embodiments.

DETAILED DESCRIPTION

[0014] Embodiments described herein relate to game play features with wagering games, and in particular to free-play wagering game redemption of cryptographic tokens for

monetary awards in a gaming environment, such as in a casino environment, and related devices, systems, and methods.

[0015] In some examples, players may access slot games or other wagering games, which may run on blockchains, using cryptographic tokens, wallets, and/or access codes. In some examples, players may play some wagering games for free, i.e., without contributing or risking real currency, and may use cryptographic tokens to convert their non-monetary credits into monetary awards. By combining blockchain technology with wagering games, such as Random Number Generator (RNG) driven games, player engagement across different channels may be increased.

[0016] In some examples, a player may obtain a crypto wallet containing one or more cryptographic tokens. In some examples, different technologies and token standards may be employed, such as ERC-20, ERC-721, ERC-777, ERC-1155, BRC-20, etc. Two-factor authentication (2FA) may be employed for enhanced security. Each of the slot-games or game-families (e.g., games connected to the a common local or wide-area jackpot) may be running on their own blockchain. The player's wallet may be connected to the slot-game prior to starting gameplay, e.g., by scanning a QR code and/or pairing with the EGM using a mobile device. In conventional free-play games, players are able to play the slot-game for free but are unable to redeem any wins but in some embodiments disclosed herein, players may be able to collect some or all of their wins as a monetary award by redeeming one or more cryptographic tokens, which in turn activates the payout mechanism.

[0017] In some examples, cryptographic tokens can be transferred, gifted, traded, and/or purchased on a secondary market, can be awarded randomly or in response to certain game events, e.g., as part of a game-win such as a bonus game, and/or can be awarded based on a challenge or task is completed (e.g., playing a game for three consecutive days in a week etc.).

[0018] In some examples, cryptographic tokens can have different characteristics, which may stack with each other, such as cosmetic features (e.g., graphics, sounds, music, game assets, etc.) and/or game features (e.g., turning different reels wild). In some examples, tokens may be stacked with other related tokens, and/or in accordance with player preferences. In some examples, stacking game features may include additional fees and/or modified odds to compensate for increased probabilities of winning outcomes, and or may reduce a number of available uses for the features. For example, a first feature may turn reel 1 wild for ten spins and a second feature may turn reel 2 wild for ten spins, but combining them such that both reel 1 and 2 are wild at the same time may only be available for three spins. In some examples, tokens may have an inactive period after being used, which may be reduced by gameplay, upgrading the token, etc.

[0019] In some examples, characteristics of each token may affect a percentage of the slot-game wins to be paid out. For example, tokens may affect characteristics of a-game such as a reel-spin speed, colors, symbol functionalities, bonus-games, math and volatility, probabilities, etc. Some tokens may be functional in certain game titles and/or product lines only, while others may be universally applicable. Tokens may have an impact on the game payout and/or on game cosmetics, and may change over time, e.g., using up game feature functionality over a period of time

while maintaining cosmetic functionality indefinitely. In some examples, tokens may be active for a predetermined time period independently of game play, for a predetermined time period after they are first redeemed, for a specific number of games, etc. For example, a token may enable monetary payouts for the next ten games played, during an entire free game bonus playoff period, for five minutes after activation, etc.

[0020] In some examples, token volatility may vary, such that the portion of the nonmonetary award that is redeemed as a monetary award can vary based on token characteristics. For example, upon token activation, a RNG with a respective volatility may determine a percentage of the total award (e.g., within a range of available percentages) that will be paid as a monetary award. For example, a high-volatility token may apply a multiplier between 0x and 5x of the total award, while a low-volatility token may apply a multiplier between 0.7x and 1.2x.

[0021] In some examples, the play session may require redemption of a token to initiate the session, with its volatility defining the characteristics of the play session, such as a maximum number of spins, a maximum wager, a maximum win, etc. For example lower-tier tokens may allow for shorter sessions, lower wins, etc., than higher-tier tokens. In some examples, ending a session early may allow remaining characteristics for the token, e.g., time and/or games remaining, to be stored on the token for later redemption by the player or another subsequent holder of the token.

[0022] Before describing these and other features in greater detail, reference is now made to FIG. 1, which illustrates a gaming system 10 including a plurality of gaming devices 100. The gaming devices 100 may be one type of a variety of different types of gaming devices, such as electronic gaming machines (EGMs), mobile gaming devices, or other devices, for example. The gaming system 10 may be located, for example, on the premises of a gaming establishment, such as a casino. The gaming devices 100, which are typically situated on a casino floor, may be in communication with each other and/or at least one central controller 40 through a data communication network 50 that may include a remote communication link. The data communication network 50 may be a private data communication network that is operated, for example, by the gaming facility that operates the gaming devices 100. Communications over the data communication network 50 may be encrypted for security. The central controller 40 may be any suitable server or computing device which includes at least one processing circuit and at least one memory or storage device. Each gaming device 100 may include a processing circuit that transmits and receives events, messages, commands or any other suitable data or signal between the gaming device 100 and the central controller 40. The gaming device processing circuit is operable to execute such communicated events, messages or commands in conjunction with the operation of the gaming device 100. Moreover, the processing circuit of the central controller 40 is configured to transmit and receive events, messages, commands or any other suitable data or signal between the central controller 40 and each of the individual gaming devices 100. In some embodiments, one or more of the functions of the central controller 40 may be performed by one or more gaming device processing circuits. Moreover, in some embodiments,

one or more of the functions of one or more gaming device processing circuits as disclosed herein may be performed by the central controller 40.

[0023] A wireless access point 60 provides wireless access to the data communication network 50. The wireless access point 60 may be connected to the data communication network 50 as illustrated in FIG. 1, and/or may be connected directly to the central controller 40 or another server connected to the data communication network 50.

[0024] A player tracking server 45 may also be connected through the data communication network 50. The player tracking server 45 may manage a player tracking account that tracks the player's gameplay and spending and/or other player preferences and customizations, manages loyalty awards for the player, manages funds deposited or advanced on behalf of the player, and other functions. Player information managed by the player tracking server 45 may be stored in a player information database 47.

[0025] As further illustrated in FIG. 1, the gaming system 10 may include a ticket server 90 that is configured to print and/or dispense wagering tickets. The ticket server 90 may be in communication with the central controller 40 through the data communication network 50. Each ticket server 90 may include a processing circuit that transmits and receives events, messages, commands or any other suitable data or signal between the ticket server 90 and the central controller 40. The ticket server 90 processing circuit may be operable to execute such communicated events, messages or commands in conjunction with the operation of the ticket server 90. Moreover, in some embodiments, one or more of the functions of one or more ticket server 90 processing circuits as disclosed herein may be performed by the central controller 40.

[0026] The gaming devices 100 communicate with one or more elements of the gaming system 10 to coordinate providing wagering games and other functionality. For example, in some embodiments, the gaming device 100 may communicate directly with the ticket server 90 over a wireless interface 62, which may be a WiFi link, a Bluetooth link, a near field communications (NFC) link, etc. In other embodiments, the gaming device 100 may communicate with the data communication network 50 (and devices connected thereto, including other gaming devices 100) over a wireless interface 64 with the wireless access point 60. The wireless interface 64 may include a WiFi link, a Bluetooth link, an NFC link, etc. In still further embodiments, the gaming devices 100 may communicate simultaneously with both the ticket server 90 over the wireless interface 66 and the wireless access point 60 over the wireless interface 64. Some embodiments provide that gaming devices 100 may communicate with other gaming devices over a wireless interface 64. In these embodiments, wireless interface 62, wireless interface 64 and wireless interface 66 may use different communication protocols and/or different communication resources, such as different frequencies, time slots, spreading codes, etc.

[0027] Embodiments herein may include different types of gaming devices. One example of a gaming device includes a gaming device 100 that can use gesture and/or touch-based inputs according to various embodiments is illustrated in FIGS. 2A, 2B, and 2C in which FIG. 2A is a perspective view of a gaming device 100 illustrating various physical features of the device, FIG. 2B is a functional block diagram that schematically illustrates an electronic relationship of

various elements of the gaming device 100, and FIG. 2C illustrates various functional modules that can be stored in a memory device of the gaming device 100. The embodiments shown in FIGS. 2A to 2C are provided as examples for illustrative purposes only. It will be appreciated that gaming devices may come in many different shapes, sizes, layouts, form factors, and configurations, and with varying numbers and types of input and output devices, and that embodiments are not limited to the particular gaming device structures described herein.

[0028] Gaming devices 100 typically include a number of standard features, many of which are illustrated in FIGS. 2A and 2B. For example, referring to FIG. 2A, a gaming device 100 (which is an EGM 160 in this embodiment) may include a support structure, housing 105 (e.g., cabinet) which provides support for a plurality of displays, inputs, outputs, controls and other features that enable a player to interact with the gaming device 100.

[0029] The gaming device 100 illustrated in FIG. 2A includes a number of display devices, including a primary display device 116 located in a central portion of the housing 105 and a secondary display device 118 located in an upper portion of the housing 105. A plurality of game components 155 are displayed on a display screen 117 of the primary display device 116. It will be appreciated that one or more of the display devices 116, 118 may be omitted, or that the display devices 116, 118 may be combined into a single display device. The gaming device 100 may further include a player tracking display 142, a credit display 120, and a bet display 122. The credit display 120 displays a player's current number of credits, cash, account balance or the equivalent. The bet display 122 displays a player's amount wagered. Locations of these displays are merely illustrative as any of these displays may be located anywhere on the gaming device 100.

[0030] The player tracking display 142 may be used to display a service window that allows the player to interact with, for example, their player loyalty account to obtain features, bonuses, comps, etc. In other embodiments, additional display screens may be provided beyond those illustrated in FIG. 2A. In some embodiments, one or more of the player tracking display 142, the credit display 120 and the bet display 122 may be displayed in one or more portions of one or more other displays that display other game related visual content. For example, one or more of the player tracking display 142, the credit display 120 and the bet display 122 may be displayed in a picture in a picture on one or more displays.

[0031] The gaming device 100 may further include a number of input devices 130 that allow a player to provide various inputs to the gaming device 100, either before, during or after a game has been played. The gaming device may further include a game play initiation button 132 and a cashout button 134. The cashout button 134 is utilized to receive a cash payment or any other suitable form of payment corresponding to a quantity of remaining credits of a credit display.

[0032] In some embodiments, one or more input devices of the gaming device 100 are one or more game play activation devices that are each used to initiate a play of a game on the gaming device 100 or a sequence of events associated with the gaming device 100 following appropriate funding of the gaming device 100. The example gaming device 100 illustrated in FIGS. 2A and 2B includes a game

play activation device in the form of a game play initiation button 132. It should be appreciated that, in other embodiments, the gaming device 100 begins game play automatically upon appropriate funding rather than upon utilization of the game play activation device.

[0033] In some embodiments, one or more input device 130 of the gaming device 100 may include wagering or betting functionality. For example, a maximum wagering or betting function may be provided that, when utilized, causes a maximum wager to be placed. Another such wagering or betting function is a repeat the bet device that, when utilized, causes the previously-placed wager to be placed. A further such wagering or betting function is a bet one function. A bet is placed upon utilization of the bet one function. The bet is increased by one credit each time the bet one device is utilized. Upon the utilization of the bet one function, a quantity of credits shown in a credit display (as described below) decreases by one, and a number of credits shown in a bet display (as described below) increases by one.

[0034] In some embodiments, as shown in FIG. 2B, the input device(s) 130 may include and/or interact with additional components, such as gesture sensors 156 for gesture input devices, and/or a touch-sensitive display that includes a digitizer 152 and a touchscreen controller 154 for touch input devices, as disclosed herein. The player may interact with the gaming device 100 by touching virtual buttons on one or more of the display devices 116, 118, 140. Accordingly, any of the above-described input devices, such as the input device 130, the game play initiation button 132 and/or the cashout button 134 may be provided as virtual buttons or regions on one or more of the display devices 116, 118, 140.

[0035] Referring briefly to FIG. 2B, operation of the primary display device 116, the secondary display device 118 and the player tracking display 142 may be controlled by a video controller 30 that receives video data from a processing circuit 12 or directly from a memory device 14 and displays the video data on the display screen. The credit display 120 and the bet display 122 are typically implemented as simple liquid crystal display (LCD) or light emitting diode (LED) displays that display a number of credits available for wagering and a number of credits being wagered on a particular game. Accordingly, the credit display 120 and the bet display 122 may be driven directly by the processing circuit 12. In some embodiments however, the credit display 120 and/or the bet display 122 may be driven by the video controller 30.

[0036] Referring again to FIG. 2A, the display devices 116, 118, 140 may include, without limitation: a cathode ray tube, a plasma display, an LCD, a display based on LEDs, a display based on a plurality of organic light-emitting diodes (OLEDs), a display based on polymer light-emitting diodes (PLEDs), a display based on a plurality of surface-conduction electron-emitters (SEDs), a display including a projected and/or reflected image, or any other suitable electronic device or display mechanism. In certain embodiments, as described above, the display devices 116, 118, 140 may include a touchscreen with an associated touchscreen controller 154 and digitizer 152. The display devices 116, 118, 140 may be of any suitable size, shape, and/or configuration. The display devices 116, 118, 140 may include flat or curved display surfaces.

[0037] The display devices 116, 118, 140 and video controller 30 of the gaming device 100 are generally configured to display one or more game and/or non-game images,

symbols, and indicia. In certain embodiments, the display devices 116, 118, 140 of the gaming device 100 are configured to display any suitable visual representation or exhibition of the movement of objects; dynamic lighting; video images; images of people, characters, places, things, and faces of cards; and the like. In certain embodiments, the display devices 116, 118, 140 of the gaming device 100 are configured to display one or more virtual reels, one or more virtual wheels, and/or one or more virtual dice. In other embodiments, certain of the displayed images, symbols, and indicia are in mechanical form. That is, in these embodiments, the display device 116, 118, 140 includes any electromechanical device, such as one or more rotatable wheels, one or more reels, and/or one or more dice, configured to display at least one or a plurality of game or other suitable images, symbols, or indicia.

[0038] The gaming device 100 also includes various features that enable a player to deposit credits in the gaming device 100 and withdraw credits from the gaming device 100, such as in the form of a payout of winnings, credits, etc. For example, the gaming device 100 may include a bill/ticket dispenser 136, a bill/ticket acceptor 128, and a coin acceptor 126 that allows the player to deposit coins into the gaming device 100.

[0039] As illustrated in FIG. 2A, the gaming device 100 may also include a currency dispenser 137 that may include a note dispenser configured to dispense paper currency and/or a coin generator configured to dispense coins or tokens in a coin payout tray.

[0040] The gaming device 100 may further include one or more speakers 150 controlled by one or more sound cards 28 (FIG. 2B). The gaming device 100 illustrated in FIG. 2A includes a pair of speakers 150. In other embodiments, additional speakers, such as surround sound speakers, may be provided within or on the housing 105. Moreover, the gaming device 100 may include built-in seating with integrated headrest speakers.

[0041] In various embodiments, the gaming device 100 may generate dynamic sounds coupled with attractive multimedia images displayed on one or more of the display devices 116, 118, 140 to provide an audio-visual representation or to otherwise display full-motion video with sound to attract players to the gaming device 100 and/or to engage the player during gameplay. In certain embodiments, the gaming device 100 may display a sequence of audio and/or visual attraction messages during idle periods to attract potential players to the gaming device 100. The videos may be customized to provide any appropriate information.

[0042] The gaming device 100 may further include a card reader 138 that is configured to read magnetic stripe cards, such as player loyalty/tracking cards, chip cards, and the like. In some embodiments, a player may insert an identification card into a card reader of the gaming device. In some embodiments, the identification card is a smart card having a programmed microchip or a magnetic strip coded with a player's identification, credit totals (or related data) and other relevant information. In other embodiments, a player may carry a portable device, such as a cell phone, a radio frequency identification tag or any other suitable wireless device, which communicates a player's identification, credit totals (or related data) and other relevant information to the gaming device. In some embodiments, money may be transferred to a gaming device through electronic funds transfer. When a player funds the gaming device, the pro-

cessing circuit determines the amount of funds entered and displays the corresponding amount on the credit or other suitable display as described above.

[0043] In some embodiments, the gaming device **100** may include an electronic payout device or module configured to fund an electronically recordable identification card or smart card or a bank or other account via an electronic funds transfer to or from the gaming device **100**.

[0044] FIG. 2B is a block diagram that illustrates logical and functional relationships between various components of a gaming device **100**. It should also be understood that components described in FIG. 2B may also be used in other computing devices, as desired, such as mobile computing devices for example. As shown in FIG. 2B, the gaming device **100** may include a processing circuit **12** that controls operations of the gaming device **100**. Although illustrated as a single processing circuit, multiple special purpose and/or general-purpose processors and/or processor cores may be provided in the gaming device **100**. For example, the gaming device **100** may include one or more of a video processor, a signal processor, a sound processor and/or a communication controller that performs one or more control functions within the gaming device **100**. The processing circuit **12** may be variously referred to as a “controller,” “microcontroller,” “microprocessor” or simply a “computer.” The processor may further include one or more application-specific integrated circuits (ASICs).

[0045] Various components of the gaming device **100** are illustrated in FIG. 2B as being connected to the processing circuit **12**. It will be appreciated that the components may be connected to the processing circuit **12** through a system bus **151**, a communication bus and controller, such as a universal serial bus (USB) controller and USB bus, a network interface, or any other suitable type of connection.

[0046] The gaming device **100** further includes a memory device **14** that stores one or more functional modules **20**. Various functional modules **20** of the gaming device **100** will be described in more detail below in connection with FIG. 2D.

[0047] The memory device **14** may store program code and instructions, executable by the processing circuit **12**, to control the gaming device **100**. The memory device **14** may also store other data such as image data, event data, player input data, random or pseudo-random number generators, pay-table data or information and applicable game rules that relate to the play of the gaming device. The memory device **14** may include random access memory (RAM), which can include non-volatile RAM (NVRAM), magnetic RAM (ARAM), ferroelectric RAM (FeRAM) and other forms as commonly understood in the gaming industry. In some embodiments, the memory device **14** may include read only memory (ROM). In some embodiments, the memory device **14** may include flash memory and/or EEPROM (electrically erasable programmable read only memory). Any other suitable magnetic, optical and/or semiconductor memory may operate in conjunction with the gaming device disclosed herein.

[0048] The gaming device **100** may further include a data storage **22**, such as a hard disk drive or flash memory. The data storage **22** may store program data, player data, audit trail data or any other type of data. The data storage **22** may include a detachable or removable memory device, including, but not limited to, a suitable cartridge, disk, CD ROM, Digital Video Disc (“DVD”) or USB memory device.

[0049] The gaming device **100** may include a communication adapter **26** that enables the gaming device **100** to communicate with remote devices over a wired and/or wireless communication network, such as a local area network (LAN), wide area network (WAN), cellular communication network, or other data communication network. The communication adapter **26** may further include circuitry for supporting short range wireless communication protocols, such as Bluetooth and/or NFC that enable the gaming device **100** to communicate, for example, with a mobile communication device operated by a player.

[0050] The gaming device **100** may include one or more internal or external communication ports that enable the processing circuit **12** to communicate with and to operate with internal or external peripheral devices, such as eye tracking devices, position tracking devices, cameras, accelerometers, arcade sticks, bar code readers, bill validators, biometric input devices, bonus devices, button panels, card readers, coin dispensers, coin hoppers, display screens or other displays or video sources, expansion buses, information panels, keypads, lights, mass storage devices, microphones, motion sensors, motors, printers, reels, Small Computer System Interface (“SCSI”) ports, solenoids, speakers, thumb drives, ticket readers, touch screens, trackballs, touchpads, wheels, and wireless communication devices. In some embodiments, internal or external peripheral devices may communicate with the processing circuit through a USB hub (not shown) connected to the processing circuit **12**.

[0051] In some embodiments, the gaming device **100** may include a sensor, such as a camera **127**, in communication with the processing circuit **12** (and possibly controlled by the processing circuit **12**) that is selectively positioned to acquire an image of a player actively using the gaming device **100** and/or the surrounding area of the gaming device **100**. In one embodiment, the camera **127** may be configured to selectively acquire still or moving (e.g., video) images and may be configured to acquire the images in either an analog, digital or other suitable format. The display devices **116**, **118**, **140** may be configured to display the image acquired by the camera **127** as well as display the visible manifestation of the game in split screen or picture-in-picture fashion. For example, the camera **127** may acquire an image of the player and the processing circuit **12** may incorporate that image into the primary and/or secondary game as a game image, symbol or indicia.

[0052] Various functional modules of that may be stored in a memory device **14** of a gaming device **100** are illustrated in FIG. 2C. Referring to FIG. 2C, the gaming device **100** may include in the memory device **14** a game module **20A** that includes program instructions and/or data for operating a hybrid wagering game as described herein. The gaming device **100** may further include a player tracking module **20B**, an electronic funds transfer module **20C**, an input device interface **20D**, an audit/reporting module **20E**, a communication module **20F**, an operating system kernel **20G** and a random number generator **20H**. The player tracking module **20B** keeps track of the play of a player. The electronic funds transfer module **20C** communicates with a back-end server or financial institution to transfer funds to and from an account associated with the player. The input device interface **20D** interacts with input devices, such as the input device **130**, as described in more detail below. The communication module **20F** enables the gaming device **100** to communicate with remote servers and other gaming

devices using various secure communication interfaces. The operating system kernel 20G controls the overall operation of the gaming device 100, including the loading and operation of other modules. The random number generator 20H generates random or pseudorandom numbers for use in the operation of the hybrid games described herein.

[0053] In some embodiments, a gaming device 100 includes a personal device, such as a desktop computer, a laptop computer, a mobile device, a tablet computer or computing device, a personal digital assistant (PDA), or other portable computing devices. In some embodiments, the gaming device 100 may be operable over a wireless network, such as part of a wireless gaming system. In such embodiments, the gaming machine may be a hand-held device, a mobile device or any other suitable wireless device that enables a player to play any suitable game at a variety of different locations. It should be appreciated that a gaming device or gaming machine as disclosed herein may be a device that has obtained approval from a regulatory gaming commission or a device that has not obtained approval from a regulatory gaming commission.

[0054] For example, referring to FIG. 2D, a gaming device 100 (which is a mobile gaming device 170 in this embodiment) may be implemented as a handheld device including a compact housing 105 on which is mounted a touchscreen display device 116 including a digitizer 152. One or more input devices 130 may be included for providing functionality of for embodiments described herein. A camera 127 may be provided in a front face of the housing 105. The housing 105 may include one or more speakers 150. In the gaming device 100, various input buttons described above, such as the cashout button, gameplay activation button, etc., may be implemented as soft buttons on the touchscreen display device 116 and/or input device 130. In this embodiment, the input device 130 is integrated into the touchscreen display device 116, but it should be understood that the input device may also, or alternatively, be separate from the display device 116. Moreover, the gaming device 100 may omit certain features, such as a bill acceptor, a ticket generator, a coin acceptor or dispenser, a card reader, secondary displays, a bet display, a credit display, etc. Credits can be deposited in or transferred from the gaming device 100 electronically.

[0055] FIG. 2E illustrates a standalone gaming device 100 (which is an EGM 160 in this embodiment) having a different form factor from the EGM 160 illustrated in FIG. 2A. In particular, the gaming device 100 is characterized by having a large, high aspect ratio, curved primary display device 116 provided in the housing 105, with no secondary display device. The primary display device 116 may include a digitizer 152 to allow touchscreen interaction with the primary display device 116. The gaming device 100 may further include a player tracking display 142, an input device 130, a bill/ticket acceptor 128, a card reader 138, and a bill/ticket dispenser 136. The gaming device 100 may further include one or more cameras 127 to enable facial recognition and/or motion tracking.

[0056] Although illustrated as certain gaming devices, such as electronic gaming machines (EGMs) and mobile gaming devices, functions and/or operations as described herein may also include wagering stations that may include electronic game tables, conventional game tables including those involving cards, dice and/or roulette, and/or other wagering stations such as sports book stations, video poker

games, skill-based games, virtual casino-style table games, or other casino or non-casino style games. Further, gaming devices according to embodiments herein may be implemented using other computing devices and mobile devices, such as smart phones, tablets, and/or personal computers, among others.

[0057] Referring now to FIG. 3, a schematic block diagram illustrates various operations for a blockchain transaction recordation according to some embodiments. As illustrated in FIG. 3, transactions 302 may occur at various blockchain nodes. In accordance with various embodiments, a hash may be created for each entry. For example, a cryptographic hash function may create a one-way, (essentially) collision free signature of the entry. The hash algorithm generates a hash. Using hashing function 304, hash values 306 of these transactions are created and added to blockchain blocks 308 that are in the blockchain data structure.

[0058] In this example, each blockchain block 308 may include one or more different types of data. For example, each blockchain block 308 may include cryptographic hash data 310 indicative of a result of the hashing function 304 associated with a common blockchain, token identifier data 312 indicative of an identity of a cryptographic token 314 associated with the block 308, common identifier data 316 indicative of the cryptographic token 314 being part of a plurality of cryptographic tokens, e.g., a defined subset of tokens based on different criteria, associated with the common blockchain, player identifier data 318 indicative of an identity of a player associated with the cryptographic token 314, transaction identifier data 320 associated with the unique transaction on the blockchain, which may include a sending address 322 associated with a provider of the cryptographic token 314 and a receiving address 324 associated with the player identifier data 318.

[0059] The cryptographic token 314 may include some or all of the data structure of the respective associated block 308 and/or may contain a block identifier 325 referring to the associated block 308, which allows the cryptographic token 314 to be securely and transparently authenticated against the blockchain. In some examples, the cryptographic token 314 may include a common identifier 326 corresponding to the common identifier data 316 and indicative of the cryptographic token 314 being part of the plurality of cryptographic tokens, and a player identifier 328 corresponding to the player identifier data 318 and indicative of a player associated with the cryptographic token 314. The cryptographic token 314 may include a unique transaction identifier 330 associated with a unique transaction on the blockchain, i.e., a unique block 308, and may include the sending address 322 and/or the receiving address 324.

[0060] For example, a new transaction for the cryptographic token may include transmitting a transaction indication of the new transaction including a sending address 322 and receiving address 324, and, based on the transaction indication, modifying the player identifier data 318 in a new block 308 and/or the player identifier 328 of the cryptographic token 314 to be indicative of a different player.

[0061] In some examples, each block 308 may further include cosmetic feature identifier data 332 associated with a cosmetic feature 334, such as a stored graphical image and/or a graphical modification of a game element of a wagering game. Each block 308 may further include game feature identifier data 336 associated with a game feature

338 for redeeming a portion of a non-monetary credit amount as a monetary award amount. Each cryptographic token **314** may be a fungible token that is fungible with other cryptographic tokens, and/or may further include a unique identifier **340**, which may correspond to the token identifier data **312** of the block **308**, such that the cryptographic token **314** is a non-fungible token that is not fungible with other cryptographic tokens.

[**0062**] As a general principle, a validation process may be performed to ensure that each new blockchain block **308** meets the criteria for inclusion into the blockchain data structure. In a blockchain configuration, there are varying consensus algorithms that can be used. For example, a private blockchain may choose an algorithm such as Practical Byzantine Fault Tolerance (PBFT). The PBFT mechanism may be useful for small networks, such as networks having fewer than about 100 nodes. Other examples include a Proof of Work (PoW) consensus algorithm and/or a Proof of Stake (POS) consensus algorithm, which may be used as the value of an underlying data block and/or value changes.

[**0063**] Referring now to FIGS. **4A-4D**, a graphical user interface (GUI) **400** is illustrated for providing a primary wagering game and a cryptographic token enabled secondary game at a gaming device **401**. As shown by FIG. **4A**, the GUI **400** may include a bet button **402**, a credit meter **404**, and a plurality of primary game elements **406** for a primary wagering game, such as a plurality of slot reels **408** and slot symbols **410** for example. In this example, the GUI **400** may further include a token indicator **412** indicative of an active cryptographic token associated with the player. In some examples, the cryptographic token may include a fungible token that is interchangeable with a plurality of cryptographic tokens and/or a unique non-fungible token, as desired.

[**0064**] In this example, the cryptographic token includes a cosmetic feature **424**, such as a stored graphical image **426** and/or a graphical modification of a game element **428** of the wagering game.

[**0065**] As shown by FIGS. **4B**, a plurality of games may be executed, with each game deducting a non-monetary wager amount **416**, indicated by the bet button **402**, from the non-monetary credit balance **418**, indicated by the credit meter **404**. A game result **420** is determined and, based on the result **420** being a winning result and the non-monetary wager amount **416**, add a non-monetary award amount **422** to the non-monetary credit balance amount **418**. In this example, the non-monetary wager amounts **416**, the non-monetary credit balance amount **418**, and the non-monetary award amounts **422** are numerical amounts corresponding to monetary amounts, e.g., dollars.

[**0066**] As shown by FIG. **4C**, the GUI **400** may display an indication of eligibility **430** to redeem the cryptographic token to award a portion of the non-monetary credit balance amount **418** to the player as a monetary award amount. For example, eligibility may be determined based on different criteria, such as a player status **432** and/or an occurrence of a game event, e.g., a bonus trigger symbol, during a predetermined time period. In this example, the eligibility indication **430** may include a predetermined time period **436**, e.g., a time limit, to redeem the token.

[**0067**] As shown in FIG. **4D**, based on an instruction to redeem the token during the eligibility period **436**, a monetary award amount **440** may be determined based on the non-monetary credit balance amount **422** and the crypto-

graphic token. In this example, a percentage **442** of the non-monetary award amount may be determined, such as a randomly determined percentage **442** from a predetermined range of percentages **444**, which may be used to determine the monetary award amount **440**. The range of percentages **444** may be based on the cryptographic token and/or different criteria, such as a stored monetary value and/or a stored game feature associated with the cryptographic token. The determined monetary award amount **446** is then awarded to the player. In some examples, redemption of the cryptographic token may disable the stored game feature, e.g., disabling redemption of the cryptographic token to convert future non-monetary awards into monetary awards. In some examples, the stored game feature of the cryptographic token may be disabled without disabling the cosmetic feature **424**.

[**0068**] FIG. **5** is a flowchart illustrating operations **500** of operations of systems/methods of independently providing a token-based secondary game at a computing device, according to some embodiments. The operations **500** may be performed by one or more processor circuits of one or more computing devices, such as any of the computing devices described herein, for example. The operations **500** may include executing, by a processor circuit of a gaming device, a plurality of games of a wagering game (Block **502**). Execution of each game may include deducting a non-monetary wager amount from a non-monetary credit balance amount (Block **504**), determining a result for the wagering game (Block **506**), and, based on the result being a winning result and the non-monetary wager amount, add a non-monetary award amount to the non-monetary credit balance amount (Block **508**).

[**0069**] After executing the plurality of games, the operations **500** may include receiving an instruction from a player of the wagering game to redeem a cryptographic token to award a portion of the non-monetary credit balance amount to the player as a monetary award amount (Block **510**). The operations **500** may further include determining, by the processor circuit, the monetary award amount based on the non-monetary credit balance amount and the cryptographic token (Block **512**). The operations **500** may further include, based on redeeming the cryptographic token, awarding, by the gaming device, the monetary award amount to the player (Block **514**).

[**0070**] Embodiments described herein may be implemented in various configurations for gaming devices **100**, including but not limited to: (1) a dedicated gaming device, wherein the computerized instructions for controlling any games (which are provided by the gaming device) are provided with the gaming device prior to delivery to a gaming establishment; and (2) a changeable gaming device, where the computerized instructions for controlling any games (which are provided by the gaming device) are downloadable to the gaming device through a data network when the gaming device is in a gaming establishment. In some embodiments, the computerized instructions for controlling any games are executed by at least one central server, central controller or remote host. In such a “thin client” embodiment, the central server remotely controls any games (or other suitable interfaces), and the gaming device is utilized to display such games (or suitable interfaces) and receive one or more inputs or commands from a player. In another embodiment, the computerized instructions for controlling any games are communicated from the central

server, central controller or remote host to a gaming device local processor and memory devices. In such a “thick client” embodiment, the gaming device local processor executes the communicated computerized instructions to control any games (or other suitable interfaces) provided to a player.

[0071] In some embodiments, a gaming device may be operated by a mobile device, such as a mobile telephone, tablet, or other mobile computing device. For example, a mobile device may be communicatively coupled to a gaming device and may include a user interface that receives user inputs that are received to control the gaming device. The user inputs may be received by the gaming device via the mobile device.

[0072] In some embodiments, one or more gaming devices in a gaming system may be thin client gaming devices and one or more gaming devices in the gaming system may be thick client gaming devices. In another embodiment, certain functions of the gaming device are implemented in a thin client environment and certain other functions of the gaming device are implemented in a thick client environment. In one such embodiment, computerized instructions for controlling any primary games are communicated from the central server to the gaming device in a thick client configuration and computerized instructions for controlling any secondary games or bonus functions are executed by a central server in a thin client configuration.

[0073] The present disclosure contemplates a variety of different gaming systems each having one or more of a plurality of different features, attributes, or characteristics. It should be appreciated that a “gaming system” as used herein refers to various configurations of: (a) one or more central servers, central controllers, or remote hosts; (b) one or more gaming devices; and/or (c) one or more personal gaming devices, such as desktop computers, laptop computers, tablet computers or computing devices, PDAs, mobile telephones such as smart phones, and other mobile computing devices.

[0074] In certain such embodiments, computerized instructions for controlling any games (such as any primary or base games and/or any secondary or bonus games) displayed by the gaming device are executed by the central server, central controller, or remote host. In such “thin client” embodiments, the central server, central controller, or remote host remotely controls any games (or other suitable interfaces) displayed by the gaming device, and the gaming device is utilized to display such games (or suitable interfaces) and to receive one or more inputs or commands. In other such embodiments, computerized instructions for controlling any games displayed by the gaming device are communicated from the central server, central controller, or remote host to the gaming device and are stored in at least one memory device of the gaming device. In such “thick client” embodiments, the at least one processor of the gaming device executes the computerized instructions to control any games (or other suitable interfaces) displayed by the gaming device.

[0075] In some embodiments in which the gaming system includes: (a) a gaming device configured to communicate with a central server, central controller, or remote host through a data network; and/or (b) a plurality of gaming devices configured to communicate with one another through a data network, the data network is an internet or an intranet. In certain such embodiments, an internet browser of the gaming device is usable to access an internet game page from any location where an internet connection is available.

In one such embodiment, after the internet game page is accessed, the central server, central controller, or remote host identifies a player prior to enabling that player to place any wagers on any plays of any wagering games. In one example, the central server, central controller, or remote host identifies the player by requiring a player account of the player to be logged into via an input of a unique username and password combination assigned to the player. It should be appreciated, however, that the central server, central controller, or remote host may identify the player in any other suitable manner, such as by validating a player tracking identification number associated with the player; by reading a player tracking card or other smart card inserted into a card reader (as described below); by validating a unique player identification number associated with the player by the central server, central controller, or remote host; or by identifying the gaming device, such as by identifying the MAC address or the IP address of the internet facilitator. In various embodiments, once the central server, central controller, or remote host identifies the player, the central server, central controller, or remote host enables placement of one or more wagers on one or more plays of one or more primary or base games and/or one or more secondary or bonus games and displays those plays via the internet browser of the gaming device.

[0076] It should be appreciated that the central server, central controller, or remote host and the gaming device are configured to connect to the data network or remote communications link in any suitable manner. In various embodiments, such a connection is accomplished via: a conventional phone line or other data transmission line, a digital subscriber line (DSL), a T-1 line, a coaxial cable, a fiber optic cable, a wireless or wired routing device, a mobile communications network connection (such as a cellular network or mobile internet network), or any other suitable medium. It should be appreciated that the expansion in the quantity of computing devices and the quantity and speed of internet connections in recent years increases opportunities for players to use a variety of gaming devices to play games from an ever-increasing quantity of remote sites. It should also be appreciated that the enhanced bandwidth of digital wireless communications may render such technology suitable for some or all communications, particularly if such communications are encrypted. Higher data transmission speeds may be useful for enhancing the sophistication and response of the display and interaction with players.

[0077] In the above description of various embodiments, various aspects may be illustrated and described herein in any of a number of patentable classes or contexts including any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof. Accordingly, various embodiments described herein may be implemented entirely by hardware, entirely by software (including firmware, resident software, micro-code, etc.) or by combining software and hardware implementation that may all generally be referred to herein as a “circuit,” “module,” “component,” or “system.” Furthermore, various embodiments described herein may take the form of a computer program product including one or more computer readable media having computer readable program code embodied thereon.

[0078] Any combination of one or more computer readable media may be used. The computer readable media may be a computer readable signal medium or a computer

readable storage medium. A computer readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer readable storage medium would include the following: a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an appropriate optical fiber with a repeater, a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any medium that can contain, or store a program for use by or in connection with an instruction execution system, apparatus, or device.

[0079] A computer readable signal medium may include a propagated data signal with computer readable program code embodied therein, for example, in baseband or as part of a carrier wave. Such a propagated signal may take any of a variety of forms, including, but not limited to, electromagnetic, optical, or any suitable combination thereof. A computer readable signal medium may be any computer readable medium that is not a computer readable storage medium and that can communicate, propagate, or transport a program for use by or in connection with an instruction execution system, apparatus, or device. Program code embodied on a computer readable signal medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber cable, radio frequency (“RF”), etc., or any suitable combination of the foregoing.

[0080] Computer program code for carrying out operations for aspects of the present disclosure may be written in any combination of one or more programming languages, including an object oriented programming language such as Java, Scala, Smalltalk, Eiffel, JADE, Emerald, C++, C#, VB.NET, Python or the like, conventional procedural programming languages, such as the “C” programming language, Visual Basic, Fortran 2003, Perl, Common Business Oriented Language (“COBOL”) 2002, PHP: Hypertext Processor (“PHP”), Advanced Business Application Programming (“ABAP”), dynamic programming languages such as Python, Ruby and Groovy, or other programming languages. The program code may execute entirely on the user’s computer, partly on the user’s computer, as a stand-alone software package, partly on the user’s computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user’s computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider) or in a cloud computing environment or offered as a service such as a Software as a Service (SaaS).

[0081] Various embodiments were described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems), devices and computer program products according to various embodiments described herein. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams,

can be implemented by computer program instructions. These computer program instructions may be provided to a processing circuit of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processing circuit of the computer or other programmable instruction execution apparatus, create a mechanism for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

[0082] These computer program instructions may also be stored in a computer readable medium that when executed can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions when stored in the computer readable medium produce an article of manufacture including instructions which when executed, cause a computer to implement the function/act specified in the flowchart and/or block diagram block or blocks. The computer program instructions may also be loaded onto a computer, other programmable instruction execution apparatus, or other devices to cause a series of operations to be performed on the computer, other programmable apparatuses or other devices to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

[0083] The flowchart and block diagrams in the figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various aspects of the present disclosure. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of code, which includes one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

[0084] The terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items and may be designated as “/”. Like reference numbers signify like elements throughout the description of the figures.

[0085] Many different embodiments have been disclosed herein, in connection with the above description and the drawings. It will be understood that it would be unduly repetitious and obfuscating to literally describe and illustrate every combination and subcombination of these embodiments. Accordingly, all embodiments can be combined in any way and/or combination, and the present specification, including the drawings, shall be construed to constitute a complete written description of all combinations and subcombinations of the embodiments described herein, and of the manner and process of making and using them, and shall support claims to any such combination or subcombination.

What is claimed is:

1. A system comprising:
 - a processor circuit; and
 - a memory coupled to the processor circuit, the memory comprising machine-readable instructions that, when executed by the processor circuit, cause the processor circuit to:
 - execute a plurality of games of a wagering game, wherein execution of each game causes the processor circuit to:
 - deduct a non-monetary wager amount from a non-monetary credit balance amount;
 - determine a result for the wagering game; and
 - based on the result being a winning result and the non-monetary wager amount, add a non-monetary award amount to the non-monetary credit balance amount;
 - after the execution of the plurality of games, receive an instruction from a player of the wagering game to redeem a cryptographic token to award a portion of the non-monetary credit balance amount to the player as a monetary award amount;
 - determine the monetary award amount based on the non-monetary credit balance amount and the cryptographic token; and
 - based on the redemption of the cryptographic token, award the monetary award amount to the player.
2. The system of claim 1, wherein the non-monetary wager amounts, the non-monetary credit balance amount, and the non-monetary award amounts comprise numerical amounts corresponding to monetary amounts.
3. The system of claim 1, wherein the determined portion of the non-monetary award amount corresponds to a determined percentage of the non-monetary credit balance amount.
4. The system of claim 3, wherein the determined percentage of the non-monetary credit balance amount is randomly determined from a predetermined range of percentages.
5. The system of claim 1, wherein the determination of the monetary award amount is further based on a stored monetary value associated with the cryptographic token.
6. The system of claim 1, wherein the determination of the monetary award amount is further based on a stored game feature associated with the cryptographic token.
7. The system of claim 6, wherein the cryptographic token further comprises a cosmetic feature.
8. The system of claim 7, wherein the cosmetic feature comprises a stored graphical image.
9. The system of claim 7, wherein the cosmetic feature comprises a graphical modification of a game element of the wagering game.

10. The system of claim 7, wherein redemption of the cryptographic token disables the stored game feature and does not disable the cosmetic feature.

11. The system of claim 1, wherein the instructions further cause the processor circuit to determine an eligibility of the player to redeem the cryptographic token, and wherein receipt of instruction to redeem a cryptographic token is further based on the determined eligibility of the player.

12. The system of claim 11, wherein the eligibility of the player is further based on the instruction to redeem the cryptographic token being received during a predetermined time period.

13. The system of claim 11, wherein the eligibility of the player is further based on a player status of the player.

14. The system of claim 11, wherein the eligibility of the player is further based on occurrence of a game event during a predetermined time period.

15. The system of claim 1, wherein the cryptographic token comprises a fungible token that is interchangeable with a plurality of cryptographic tokens.

16. The system of claim 1, wherein the cryptographic token comprises a unique non-fungible token.

17. A gaming device comprising:

- a display device;
- an input device;
- a processor circuit; and
- a memory coupled to the processor circuit, the memory comprising machine-readable instructions that, when executed by the processor circuit, cause the processor circuit to:
 - receive, via the input device, a plurality of non-monetary wagers for a wagering game;
 - based on the plurality of non-monetary wagers, execute a plurality of games of a wagering game, wherein execution of each game causes the processor circuit to:
 - deduct a non-monetary wager amount of the non-monetary wager from a non-monetary credit balance amount;
 - display, by the display device, a result for the wagering game; and
 - based on the result being a winning result and the non-monetary wager amount, add a non-monetary award amount to the non-monetary credit balance amount;
 - after the execution of the plurality of games, receive, at the input device, an instruction from a player of the wagering game to redeem a cryptographic token to award a portion of the non-monetary credit balance amount to the player as a monetary award amount;
 - determine the monetary award amount based on the non-monetary credit balance amount and the cryptographic token; and
 - based on the redemption of the cryptographic token, display, by the display device, an indication of the monetary award amount, and award the monetary award amount to the player.

18. The gaming device of claim 17, wherein the non-monetary wager amounts, the non-monetary credit balance amount, and the non-monetary award amounts comprise numerical amounts corresponding to monetary amounts.

- 19.** A method comprising:
executing, by a processor circuit of a gaming device, a plurality of games of a wagering game, wherein execution of each game comprises:
deducting a non-monetary wager amount from a non-monetary credit balance amount;
determining a result for the wagering game; and
based on the result being a winning result and the non-monetary wager amount, adding a non-monetary award amount to the non-monetary credit balance amount;
after executing the plurality of games, receiving an instruction from a player of the wagering game to redeem a cryptographic token to award a portion of the non-monetary credit balance amount to the player as a monetary award amount;
determining, by the processor circuit, the monetary award amount based on the non-monetary credit balance amount and the cryptographic token; and
based on redeeming the cryptographic token, awarding, by the gaming device, the monetary award amount to the player.
- 20.** The method of claim **19**, wherein the non-monetary wager amounts, the non-monetary credit balance amount, and the non-monetary award amounts comprise numerical amounts corresponding to monetary amounts.

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